

Introducing InfiniBand at ACEnet

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What is InfiniBand?

InfiniBand is a low-latency high-bandwidth networking technology. It can be used to connect compute nodes with each other, similarly to Ethernet. Infiniband is often used in high-performance computing, and it can provide significant performance improvements for many applications.

ACEnet installation

In the end of 2009, ACEnet acquired and deployed the InfiniBand interconnect at the Fundy (UNB) and Placentia (MUN) clusters. The installed Mellanox HCA cards and high-capacity Voltaire switches are capable of delivering the 4X DDR signalling rate, or 20 Gbit/s of theoretical throughput.

The InfiniBand network at ACEnet is used for the MPI messaging, and is not utilized for network storage data transfers.

Before this installation, only the Mahone (SMU) cluster was equipped with the high-performance interconnect – Myrinet.

What should I do to start using InfiniBand?

You do not have to do anything. Provided that an application you are running is an MPI code compiled with the Open MPI library (which is default at ACEnet), any available fast interconnect (e.g. InfiniBand) will be automatically activated. Hence, any MPI software package provided by ACEnet is InfiniBand-ready, as well as any user's application compiled with Open MPI.

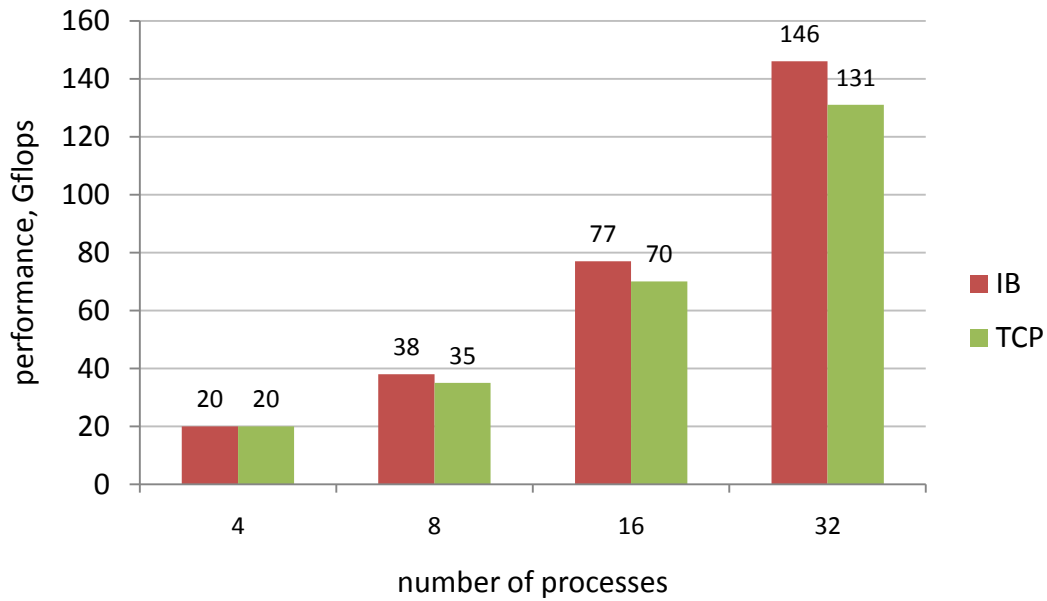
InfiniBand vs Ethernet

Below are some performance comparison results for several applications run over InfiniBand (IB) and 1Gbit Ethernet (TCP). These applications represent different scientific fields: chemistry, astrophysics and oceanography. The input parameters used are those from real scientific calculations and were provided by ACEnet users. There are also comparison data for the two de facto performance benchmarks.

All the tests were performed at Placentia on 8 dual-socket dual-core X2200 nodes running at 3GHz. Every test was run several times to verify consistency of the results.

HPL

HPL is the parallel implementation of the Linpack benchmark. It is used as performance measure to rank computer systems in the TOP500 project.

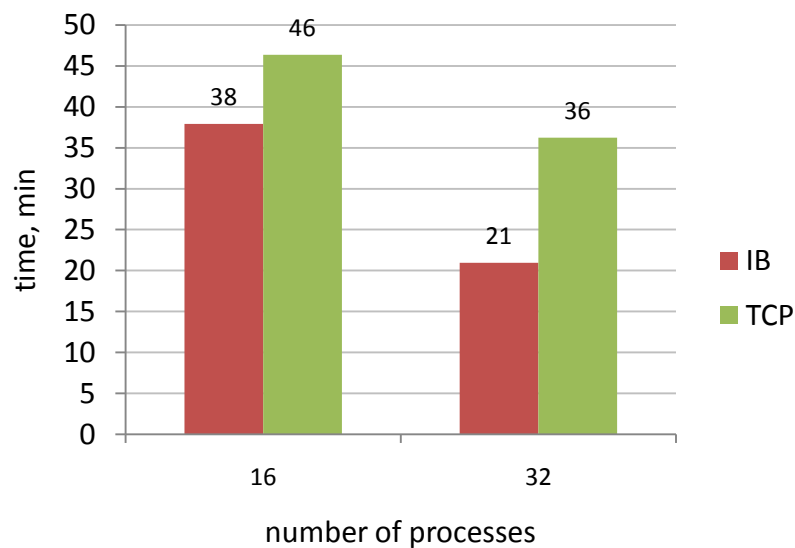


As you can see, the HPL benchmark does not depend much on the interconnect type, and the performance improvement for a 32-proc job is just above 11%.

Please note that the HPL test was not purposely tuned to get the best performance number possible. It also takes different input data for jobs of different sizes. Hence, the graph should only be viewed as a comparison between InfiniBand and Ethernet, and not as a system performance benchmark.

NEMO

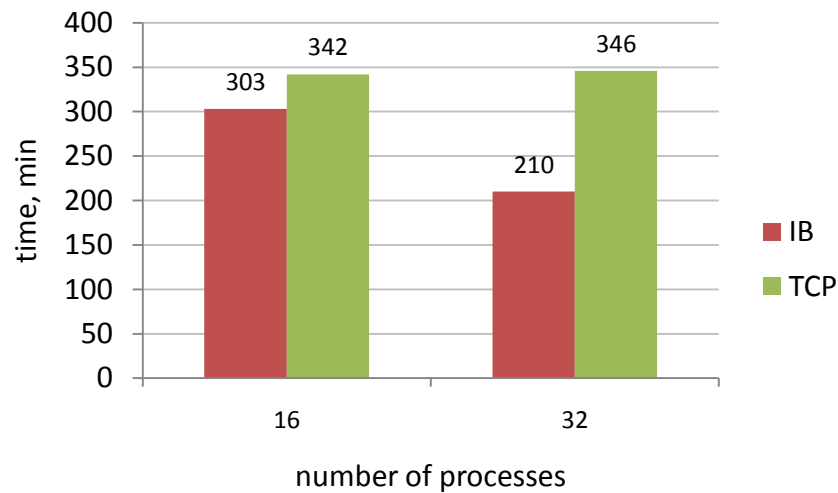
NEMO is a state-of-the-art modeling framework for oceanographic research, operational oceanography seasonal forecast and climate studies.



Not only is the job running over InfiniBand faster (21% for a 16-proc, 71% for a 32-proc), it also scales much better with the growing number of processes: a 81% improvement for the InfiniBand job vs. 28% for Ethernet.

Enzo

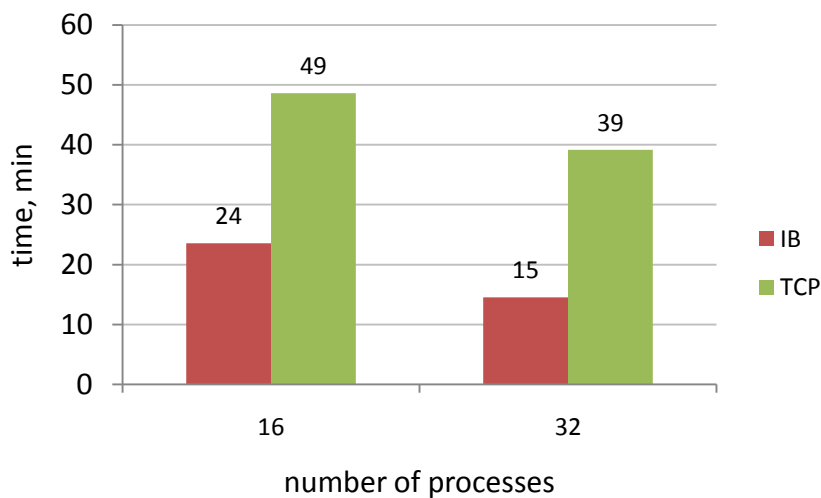
Enzo is an adaptive mesh refinement, grid-based hybrid code (hydro + N-Body) which is designed to do simulations of cosmological structure formation.



Interestingly, doubling the number of processes does not improve performance at all for a job running over Ethernet, most likely due to insufficient bandwidth of the Ethernet. At the same time, if run over InfiniBand, the 32-proc job is 44% faster than the 16-proc one.

CPMD

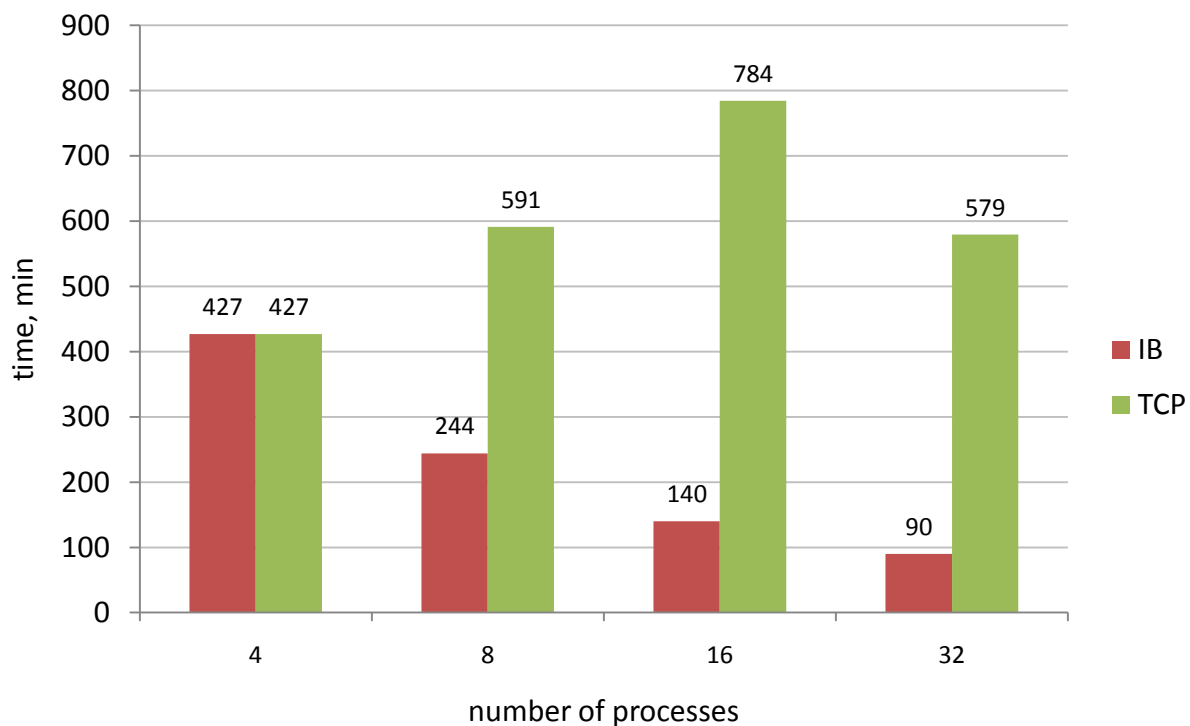
The CPMD code is a parallelized plane wave/pseudopotential implementation of Density Functional Theory, particularly designed for ab-initio molecular dynamics.



This is a good illustration of how using a fast interconnect can improve both performance and scalability delivering stellar overall performance improvement with the same amount of compute power. While for a 16-proc job, a mere switch from Ethernet to InfiniBand doubles the performance, for a larger 32-proc job, the performance improves by a factor of 2.6.

Gromacs

Gromacs is a molecular dynamics package primarily designed for biomolecular systems such as proteins and lipids.

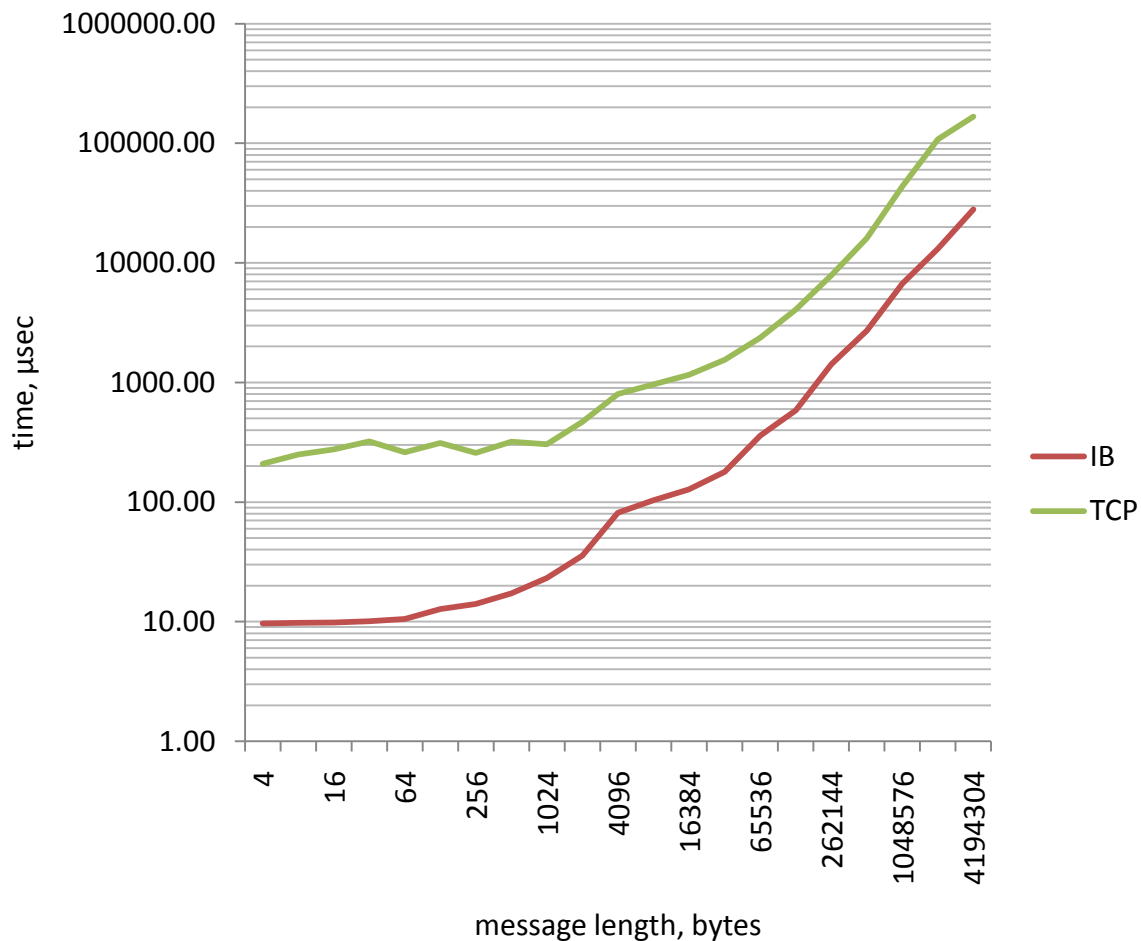


In this case, the Ethernet cannot provide better performance for a larger job – the code is actually performing worse running on more compute cores (with more processes), most likely due to substantially increased amount of network traffic.

In this situation, if someone wanted to run a more complex simulation, they would be forced to use just 4 cores, not being able to take advantage of additional compute power, making it unlikely to get results in a reasonable amount of time. Contrary, the job running over InfiniBand is demonstrating consistent performance improvement with the increasing compute power.

IMB

IMB is an MPI performance benchmark suite. The figure below shows the results for the Allreduce benchmark from the suite, which was run as a 16-proc job. This example should give an idea about overall advantage of using InfiniBand over Ethernet.



Conclusions

- The InfiniBand network provides more bandwidth and lower latency than 1Gbit Ethernet. It's used exclusively for the MPI messaging, and every compute node is connected directly to the same InfiniBand switch within a cluster. Most likely your MPI application will benefit immediately from being run over InfiniBand instead of Ethernet.
- With no network congestion and other bottlenecks associated with the Ethernet architecture, the scalability of your application is very likely to improve as well.
- Better performance and scalability will allow you to run larger and more complex simulations, which will be completed in a reasonable amount of time.
- There is nothing you have to do in order to make use of InfiniBand; no need to recompile or reconfigure anything. As long as your MPI application is built with the Open MPI library (default at ACEnet), the high-speed network (InfiniBand or Myrinet) will be used automatically, if available. Naturally, all MPI applications provided by ACEnet are InfiniBand-ready.